Neurodevelopmental Disorder With Absent Speech and Movement and Behavioral Abnormalities (NEDASMB)

A comprehensive disease-characteristics report for knowledge-base curation

Disease: Neurodevelopmental Disorder With Absent Speech and Movement and Behavioral Abnormalities (NEDASMB) **MONDO ID:** MONDO:0859519 · **OMIM phenotype:** #620270 · **Gene:** *UBE3C* (MIM 614454) **Category:** Mendelian (autosomal recessive)

Summary

Neurodevelopmental Disorder With Absent Speech and Movement and Behavioral Abnormalities (NEDASMB) is an ultra-rare Mendelian disorder caused by **biallelic (autosomal-recessive) loss-of-function variants in *UBE3C*** (chromosome 7q36.3), and catalogued as **OMIM #620270 / MONDO:0859519 / MedGen C5830319 (UID 1840955)**. It was first delineated in 2023 by Faqeih and colleagues, who used a paralog-directed candidate-gene approach — reasoning that because *UBE3C* is the paralog of the Angelman-syndrome gene *UBE3A*, its loss might produce an Angelman-like phenotype. They identified three patients from two families with an Angelman-like syndrome carrying two loss-of-function *UBE3C* variants; patient RNA studies confirmed the loss-of-function effect [PMID: 36401616].

Clinically, affected individuals present from infancy with a static (non-progressive) encephalopathy comprising **severe global developmental delay/intellectual disability, absent speech, a movement disorder (ataxia, dystonia, tremor), seizures, and neurobehavioral abnormalities** (autistic features, hyperactivity, aggression, sleep disturbance), together with variable brain-imaging findings such as a thin corpus callosum and cerebellar hypoplasia. Mechanistically, *UBE3C* encodes a **proteasome-associated HECT-type E3 ubiquitin ligase** (UniProt Q15386, 1083 aa) that assembles K48- and K29-linked polyubiquitin chains and enhances proteasome processivity. Its loss is predicted to disrupt neuronal protein homeostasis (proteostasis) during cortical circuit assembly — placing

NEDASMB conceptually alongside Angelman syndrome (*UBE3A*) and Kaufman oculocerebrofacial syndrome (*UBE3B*) as a "UBE3-family" neurodevelopmental proteostasis disorder.

Because the disorder was described from a single small cohort, most epidemiological, natural-history, prognostic, and disease-specific molecular parameters remain undefined. There is **no disease-specific treatment**; management is entirely supportive/symptomatic, and diagnosis rests on **exome or genome sequencing** after excluding *UBE3A*/15q11–q13 (Angelman) defects. This report consolidates the confirmed identifiers, genetics, mechanism, phenotype, and knowledge gaps to support a curated disease entry, and flags every place where evidence is currently absent.

Key Findings

Finding 1 — NEDASMB is caused by biallelic loss-of-function variants in *UBE3C*

NEDASMB (MONDO:0859519, OMIM #620270) is caused by **biallelic loss-of-function (LoF) variants in *UBE3C*** (HGNC:16803; NCBI GeneID 9690; ENSG00000009335; chr7q36.3, GRCh38 chr7:157,138,916–157,269,370). The disorder was defined by a **paralog-directed candidate-gene approach**: because *UBE3C* is the paralog of the Angelman-syndrome E3 ligase gene *UBE3A*, the authors screened patients with an Angelman-like clinical picture and detected two LoF *UBE3C* variants in three patients from two families. Patient RNA studies confirmed the loss-of-function consequence, and inheritance is autosomal recessive [PMID: 36401616].

> "In 3 patients from 2 families with Angelman-like syndrome, paralog-directed candidate gene approach detected 2 LoF variants in the other candidate E3 ligase gene, UBE3C, a paralog of the Angelman syndrome E3 ligase gene, UBE3A." [PMID: 36401616]
>
> "HECTD4 and UBE3C are novel biallelic rare disease genes, expand the association of the other HECT E3 ligase group with neurodevelopmental syndromes, and could explain some of the missing heritability in patients with a suggestive clinical diagnosis of Angelman syndrome." [PMID: 36401616]

The gene is highly constrained in population databases (gnomAD: pLI = 1.0, LOEUF/oe_lof upper = 0.42, missense-z = 3.17), consistent with strong intolerance to loss-of-function — a genomic signature expected for a gene whose complete biallelic loss produces severe neurodevelopmental disease.

Finding 2 — *UBE3C* is a proteasome-associated HECT E3 ligase essential for neuronal proteostasis and cortical development

UBE3C belongs to the **"other" subfamily of HECT-type E3 ubiquitin ligases**. Its catalytic HECT domain (aa 744–1083) adopts an open, L-shaped bilobed conformation and autoubiquitinates (major site Lys903); the last three C-terminal residues are essential for catalysis [PMID: 32039437]. UBE3C assembles **K48- and K29-linked polyubiquitin chains** [PMID: 25752577] and participates in proteasome processivity, innate immunity, and cancer metastasis.

> "The UBE3C E3 ligase is a member of the 'other' subfamily HECT and influences several crucial cellular processes, including innate immunity, proteasome processivity, and cancer metastasis." [PMID: 32039437]
>
> "the human HECT E3 ligases UBE3C and AREL1 assemble K48/K29- and K11/K33-linked Ub chains" [PMID: 25752577]

Critically, *UBE3C* is a **brain-expressed member of the UBE3 paralog family** (alongside *UBE3A*/Angelman syndrome and *UBE3B*/Kaufman oculocerebrofacial syndrome), and the family's E3-ligase activity is required for cortical circuit assembly and higher cognition [PMID: 33182779]. A separate *UBE3C*-reducing structural variant causes distal hereditary motor neuropathy (DHMN1) with *C. elegans* synaptic-transmission deficits and heat-stress susceptibility, further implicating protein homeostasis (P 36380488 , from investigation notes).

> "uninterrupted action of UBE3 ligases is a sine qua non for cortical circuit assembly and higher cognitive functions of the neocortex" [PMID: 33182779]

Finding 3 — Clinical phenotype: severe global developmental delay, absent speech, movement disorder, seizures, behavioral abnormalities (Angelman-like)

Faqeih et al. described patients with "syndromic neurodevelopmental, seizure, and movement disorders and neurobehavioral phenotypes" in an Angelman-like presentation [PMID: 36401616]. HPO annotations for OMIM:620270 (via Monarch, n = 23 features) span developmental, motor, seizure, and behavioral domains.

> "Chromosomal analysis and exome sequencing were used to identify the genetic causes in 10 patients from 7 unrelated families with syndromic neurodevelopmental, seizure, and movement disorders and neurobehavioral phenotypes." [PMID: 36401616]

Onset is congenital/infantile, and the course is chronic and non-progressive (static encephalopathy). The phenotype closely mirrors Angelman syndrome, which is characterized by "severe mental retardation, epilepsy, absent speech, dysmorphic facial features, and a characteristic behavioral phenotype" [PMID: 17848870], supporting the paralog-driven clinical overlap.

HPO term	Phenotype	Domain
HP:0001263	Global developmental delay	Neurodevelopmental
HP:0010864	Severe intellectual disability	Neurodevelopmental
HP:0002300	Mutism / absent speech	Speech
HP:0001270	Motor delay	Motor
HP:0001252	Hypotonia	Motor
HP:0001251	Ataxia	Movement disorder
HP:0002451	Limb dystonia	Movement disorder
HP:0001337	Tremor	Movement disorder
HP:0000729	Autistic behavior	Behavioral
HP:0000752	Hyperactivity	Behavioral
HP:0000718	Aggressive behavior	Behavioral
HP:0002360	Sleep disturbance	Behavioral
HP:0011968	Feeding difficulties	Systemic
HP:0033725	Thin corpus callosum	Brain imaging
HP:0001321	Cerebellar hypoplasia	Brain imaging
HP:0000365	Hearing impairment	Sensory

Finding 4 — UBE3C/Hul5 is a proteasome processivity factor; mouse knockouts show neurobehavioral phenotypes

UBE3C (yeast ortholog **Hul5**) is one of ~5 ubiquitin ligases associated with the mammalian 26S proteasome; it enhances proteasomal processivity and cycles on/off the proteasome with the deubiquitinase Usp14/Ubp6 in response to ubiquitinated-substrate supply [PMID: 28396413].

> "the ubiquitin ligase, Ube3c/Hul5, which enhances proteasomal processivity" [PMID: 28396413]

Under proteotoxic stress (heat shock, arsenite, proteasome inhibition), Ube3c/Hul5 selectively polyubiquitinates the proteasome ubiquitin-receptor subunit **Rpn13**, reducing the proteasome's capacity to degrade ubiquitin conjugates — an autoinhibitory / stress-biomarker mechanism [PMID: 24811749].

> "Rpn13 becomes extensively and selectively poly-ubiquitinated by the proteasome-associated ubiquitin ligase, Ube3c/Hul5" [PMID: 24811749]

IMPC mouse *Ube3c*-null lines (mouse gene ID 100763) show neurobehavioral abnormalities — decreased prepulse inhibition, decreased locomotor/vertical activity, decreased exploration in a new environment, and abnormal gait — plus metabolic, cardiac, and immune findings, providing a model that partially recapitulates the human movement/behavioral phenotype.

Finding 5 — Protein features and variant landscape

UniProt Q15386 (UBE3C_HUMAN, 1083 aa): an E3 ubiquitin-protein ligase that catalyzes Lys-29- and Lys-48-linked polyubiquitin chains, accepts ubiquitin from the E2 enzyme **UBE2D1** as a thioester, and associates with the 26S proteasome to elongate ubiquitin chains on bound substrates. **ClinVar** contains ~257 *UBE3C* variant records; isolated single-nucleotide variants are predominantly **Uncertain Significance (VUS)** with a few Likely benign, whereas the "Pathogenic" entries at this locus are large 7q36 copy-number changes (contiguous-gene events) rather than isolated *UBE3C* point mutations. The disease-defining biallelic LoF alleles from

P 36401616

are private/rare and not yet broadly deposited.

Finding 6 — Pathway context and expression

Reactome maps UBE3C (Q15386) to **R-HSA-983168** "Antigen processing: Ubiquitination & Proteasome degradation" within the Class-I-MHC-mediated antigen-processing superpathway; relevant KEGG pathways are **hsa03050 (Proteasome)** and **hsa04120 (Ubiquitin-mediated proteolysis)**. UBE3C shows **low tissue specificity** (broad/ubiquitous expression, Human Protein Atlas) and is brain-expressed [PMID: 33182779]. No NEDASMB-specific transcriptomic, proteomic, metabolomic, or single-cell datasets exist; the only patient functional data are the RNA studies confirming the LoF effect [PMID: 36401616].

Finding 7 — OMIM #620270 is the *UBE3C*-specific phenotype entry

NCBI Gene → OMIM linkage for *UBE3C* (GeneID 9690, 7q36.3) returns two records: **phenotype MIM 620270** (Neurodevelopmental disorder with absent speech and movement and behavioral abnormalities) and **gene MIM 614454** (*UBE3C*). MedGen concept **C5830319 (UID 1840955)** corresponds to the same phenotype. This confirms MONDO:0859519 = OMIM:620270 is the *UBE3C*-specific disorder, distinct from the paralog *HECTD4* phenotype reported in the same paper [PMID: 36401616].

Section-by-Section Report

1. Disease Information

- **Overview:** NEDASMB is an ultra-rare autosomal-recessive neurodevelopmental disorder in which biallelic loss of the proteasome-associated E3 ubiquitin ligase *UBE3C* produces an Angelman-like syndrome of severe developmental delay, absent speech, movement disorder, seizures, and behavioral abnormalities.
- **Key identifiers:** OMIM phenotype **#620270**; gene MIM **614454**; **MONDO:0859519**; **MedGen C5830319 (UID 1840955)**; **UMLS C5830319**. No dedicated Orphanet or ICD-10/ICD-11 code exists yet. MeSH lacks a specific descriptor; the closest umbrella term is "Neurodevelopmental Disorders" (MeSH D065886).
- **Synonyms / alternative names:** NEDASMB; UBE3C-related neurodevelopmental disorder; Angelman-like syndrome due to UBE3C deficiency (informal).
- **Information source:** Aggregated disease-level resources (OMIM, MONDO, MedGen) built from a single primary clinical/genetic report [PMID: 36401616], i.e., individual-patient data from a small cohort rather than EHR-scale or registry data.

2. Etiology

- **Causal factor:** Purely genetic — biallelic (homozygous or compound-heterozygous) loss-of-function variants in *UBE3C* [PMID: 36401616]. No environmental, infectious, or toxic cause is implicated.
- **Genetic risk factors:** The causal variants are the *UBE3C* LoF alleles themselves. No susceptibility loci or modifier genes have been reported. As an autosomal-recessive disorder, **consanguinity** is a demographic risk factor, though this was not quantified.

- **Environmental risk / protective factors:** None identified — not applicable to a monogenic recessive disorder.
- **Gene–environment interactions:** None reported. *In vitro* and model-organism data show UBE3C activity is modulated by **proteotoxic stress** (heat shock, arsenite, proteasome inhibition) [PMID: 24811749], suggesting a theoretical stress-dependent modifier axis, but this has not been demonstrated in patients.

3. Phenotypes

The phenotype is Angelman-like and dominated by CNS features (see Finding 3 table for HPO terms). **Phenotype types** include behavioral changes (autistic behavior, hyperactivity, aggression, sleep disturbance), clinical/neurological signs (hypotonia, ataxia, dystonia, tremor, seizures), developmental manifestations (global developmental delay, severe intellectual disability, absent speech, motor delay), and structural brain findings (thin corpus callosum, cerebellar hypoplasia). **Onset** is congenital/infantile; **severity** is severe; **progression** is static (non-progressive encephalopathy); **frequency among affected individuals** cannot be reliably quantified given the tiny cohort (n = 3 for the *UBE3C*-specific phenotype). **Quality-of-life impact** is presumed profound (non-verbal, dependent for activities of daily living), but no formal QoL instrument data (EQ-5D, SF-36, PROMIS) exist for this disorder.

4. Genetic / Molecular Information

- **Causal gene:** *UBE3C* (HGNC:16803; NCBI GeneID 9690; MIM 614454; ENSG00000009335; 7q36.3).
- **Pathogenic variants:** Biallelic LoF alleles (e.g., nonsense, frameshift, splice — consistent with the LoF mechanism confirmed by patient RNA studies) [PMID: 36401616]. In ClinVar, isolated *UBE3C* SNVs are mostly VUS; large 7q36 CNVs are catalogued as pathogenic contiguous-gene events. Population allele frequency of the disease alleles is very low/private (gnomAD constraint pLI = 1.0).
- **Origin:** Germline (autosomal recessive). No somatic role in this disease.
- **Functional consequence:** Loss of function (reduced/absent proteasome-associated E3 ligase activity).
- **Modifier genes / epigenetics / chromosomal abnormalities:** None specifically defined for NEDASMB. Note that *UBE3A* (the paralog) is imprinted, but there is no evidence *UBE3C* is imprinted; NEDASMB follows classic biallelic recessive genetics rather than parent-of-origin effects.

5. Environmental Information

Not applicable — NEDASMB is a monogenic recessive disorder with **no known environmental, lifestyle, or infectious contributors**. Cellular studies show UBE3C responds to proteotoxic stressors [PMID: 24811749], but no environmental trigger has been linked to disease onset or severity in patients.

6. Mechanism / Pathophysiology

Molecular pathway: Ubiquitin–proteasome system (UPS). UBE3C is a proteasome-associated HECT E3 ligase that assembles **K48- and K29-linked polyubiquitin chains** [PMID: 25752577] using E2 UBE2D1, and enhances proteasome processivity as substrates are threaded into the 26S proteasome [PMID: 28396413]. Relevant pathways: Reactome R-HSA-983168; KEGG hsa03050 and hsa04120.

Cellular process: Regulated protein degradation / protein quality control (proteostasis). Loss of UBE3C is predicted to impair clearance of ubiquitinated neuronal substrates during a period when the neocortex requires "uninterrupted action of UBE3 ligases … for cortical circuit assembly" [PMID: 33182779].

Protein dysfunction: Loss-of-function — reduced catalytic output of the HECT domain (catalytic region aa 744–1083; C-terminal residues essential) [PMID: 32039437].

Proposed causal chain:

```
Biallelic UBE3C LoF variants
        │
        ▼
Loss of proteasome-associated E3 ligase activity
(reduced K48/K29 polyubiquitination; reduced proteasome processivity)
        │
        ▼
Impaired neuronal protein homeostasis (proteostasis)
during cortical & cerebellar circuit assembly
        │
        ▼
Aberrant neurodevelopment
(thin corpus callosum, cerebellar hypoplasia)
        │
        ▼
Clinical phenotype: severe DD/ID, absent speech,
movement disorder, seizures, behavioral abnormalities
```

Upstream event = loss of UBE3C enzymatic function; downstream events = disrupted proteostasis → aberrant circuit assembly → clinical manifestations. **Cell types involved:** cortical and cerebellar neurons (CL:0000540 neuron; CL:0000121 Purkinje cell — inferred from cerebellar hypoplasia). **GO terms:** GO:0016567 protein ubiquitination; GO:0000209 protein polyubiquitination; GO:0043161 proteasome-mediated ubiquitin-dependent protein catabolic process; GO:0007420 brain development; GO:0021987 cerebral cortex development. **CHEBI:** ubiquitin conjugate / ATP (cofactors for proteasomal degradation).

Immune / metabolic / other: UBE3C also functions in innate immunity (HECT-ligase regulation of type-I IFN) and cancer metastasis [PMID: 32039437; P 21167755], but these are not established components of NEDASMB pathophysiology. No disease-specific metabolomic, proteomic, transcriptomic, or single-cell profiling of patient tissue exists.

7. Anatomical Structures Affected

- **Organ / system:** Central nervous system (nervous system; UBERON:0001016). Primary site = brain (UBERON:0000955), with imaging evidence pointing to the **corpus callosum** (UBERON:0002336; thin) and **cerebellum** (UBERON:0002037; hypoplasia). Sensory involvement includes hearing impairment (auditory pathway).
- **Tissue / cell level:** Nervous tissue; cortical and cerebellar neurons (CL:0000540; CL:0000121 Purkinje cell).
- **Subcellular level:** Cytoplasm and 26S proteasome (GO:0000502 proteasome complex; GO:0005829 cytosol); UBE3C associates with the proteasome and cycles on/off it [PMID: 28396413].
- **Lateralization:** Bilateral/symmetric CNS involvement (consistent with a diffuse genetic encephalopathy).

8. Temporal Development

- **Onset:** Congenital/infantile; developmental delay evident from early infancy.
- **Onset pattern:** Chronic/insidious (developmental).
- **Progression:** Static (non-progressive) encephalopathy; no evidence of regression in the reported cohort. Seizures and behavioral features persist across childhood.
- **Duration:** Chronic, lifelong.

- **Critical periods:** Prenatal/early-postnatal cortical and cerebellar circuit assembly is the mechanistically implicated vulnerable window [PMID: 33182779]; there is no defined therapeutic window because no disease-modifying therapy exists.

9. Inheritance and Population

- **Inheritance:** Autosomal recessive (biallelic LoF) [PMID: 36401616].
- **Epidemiology:** Prevalence and incidence are **undefined** (ultra-rare; ~3 patients reported for the *UBE3C*-specific phenotype). Not in Orphanet/GBD with a prevalence estimate.
- **Penetrance / expressivity:** Presumed high penetrance for biallelic LoF, but formal penetrance and the range of expressivity cannot be estimated from n = 3.
- **Anticipation:** Not applicable (not a repeat-expansion disorder).
- **Founder effects / consanguinity:** Recessive inheritance implies consanguinity or shared ancestry may raise risk; no founder allele has been formally established.
- **Carrier frequency:** Not established; expected to be very low given gnomAD constraint.
- **Sex ratio / age / geographic distribution:** Not established.

10. Diagnostics

- **Primary diagnostic modality: Exome sequencing (WES)** or **genome sequencing (WGS)**, which identified the causal variants; chromosomal analysis was also used in the discovery cohort [PMID: 36401616]. Because the phenotype overlaps Angelman syndrome, the recommended workup first **excludes *UBE3A*/15q11–q13 defects** (methylation analysis, UBE3A sequencing, chromosomal microarray) before attributing disease to *UBE3C*.
- **Confirmatory testing:** Bi-allelic *UBE3C* LoF variants with segregation; patient **RNA studies** confirmed the LoF effect and are a useful functional adjunct [PMID: 36401616].
- **Imaging:** Brain MRI may reveal thin corpus callosum and cerebellar hypoplasia (supportive, not specific).
- **Electrophysiology:** EEG for seizure characterization (as in Angelman-like disorders, though no NEDASMB-specific EEG signature is defined).
- **Biomarkers / omics diagnostics:** No validated blood, CSF, metabolomic, or proteomic biomarker. No newborn-screening test.
- **Clinical criteria / differential diagnosis:** No formal diagnostic criteria. Differential diagnosis includes **Angelman syndrome (*UBE3A*)**, and other severe DD/ID-absent-speech-behavioral syndromes such as *MEF2C* haploinsufficiency [PMID: 27255693], *ACTL6B*-related disorder [PMID: 39275948], *CTNNB1* syndrome [PMID: 36293418], *KMT5B*-related disorder

[PMID: 37927187], and *SYT1*-related disorder [PMID: 38058756]. Genetic testing distinguishes these.

11. Outcome / Prognosis

- **Survival / mortality:** No mortality or life-expectancy data reported. The disorder appears to be a chronic static encephalopathy compatible with survival into at least adolescence/adulthood, but this is not quantified.
- **Morbidity / disability:** Profound — non-verbal, severe intellectual disability, motor impairment, dependence for daily activities; seizures and behavioral disturbance add morbidity.
- **Recovery / prognostic factors / prognostic biomarkers:** None defined. Given static course and supportive-only management, meaningful recovery is not expected. No QoL instrument data exist.

12. Treatment

- **Disease-modifying therapy: None exists.** Management is entirely **supportive and symptomatic**.
- **Symptomatic pharmacotherapy (by analogy to Angelman-like disorders):** Anti-seizure medications for epilepsy; agents for sleep disturbance; behavioral/ADHD management as needed. No pharmacogenomic guidance is specific to *UBE3C*. (NCIT: C15607 Anticonvulsant Therapy; C15632 Supportive Care.)
- **Rehabilitative / supportive care:** Physical therapy, occupational therapy, speech/AAC (augmentative and alternative communication) for non-verbal individuals, feeding support for feeding difficulties. AAC design research in intellectual and developmental disabilities is relevant to communication support [PMID: 29710313]. (NCIT: C15315 Physical Therapy; C15258 Rehabilitation Therapy.)
- **Advanced / experimental therapeutics:** None reported; no gene therapy, ASO, or targeted therapy in trials for NEDASMB. No ClinicalTrials.gov entries were identified for this specific disorder.

13. Prevention

- **Primary prevention:** Not applicable (genetic disorder). **Genetic counseling** for at-risk families is the principal preventive intervention: recessive recurrence risk of 25% per pregnancy for carrier couples.

- **Screening / early detection: Carrier screening** and, where a familial variant is known, **cascade testing**, **prenatal diagnosis**, or **preimplantation genetic testing (PGT-M)** are options. No population-level or newborn screening exists.
- **Public-health / behavioral / immunization measures:** Not applicable.

14. Other Species / Natural Disease

- **Orthologs:** Yeast *HUL5* (functional ortholog, processivity factor) [PMID: 28396413]; mouse *Ube3c* (NCBI Gene 100763); *C. elegans* ortholog shows synaptic-transmission phenotypes when UBE3C is reduced (P 36380488, investigation notes).
- **Natural disease in other species:** No naturally occurring *UBE3C*-related disease is catalogued in OMIA for companion animals or wildlife.
- **Comparative biology:** UBE3C function (proteasome-associated processivity, K48/K29 chain assembly) is conserved from yeast to human, supporting cross-species mechanistic study.
- **Zoonotic potential:** Not applicable.

15. Model Organisms

Model	System	Phenotype recapitulation	Reference
Mouse *Ube3c*-null (IMPC, gene 100763)	Mammalian, knockout	Neurobehavioral: ↓prepulse inhibition, ↓locomotor/vertical activity, ↓exploration, abnormal gait; plus metabolic/cardiac/immune findings	IMPC
C. elegans (UBE3C reduction)	Invertebrate	Synaptic-transmission deficits; heat-stress susceptibility (proteostasis)	P 36380488
S. cerevisiae hul5Δ	Yeast	Loss of proteasome processivity; stress-dependent Rpn13 ubiquitination biology	P 28396413; P 24811749

- **Genetic model types available:** Knockout mouse (IMPC); worm and yeast loss-of-function.

- **Model limitations:** No published mouse model engineered with a patient-specific biallelic LoF allele or assessed against the full human cognitive/speech/seizure phenotype; behavioral endophenotypes only partially map to human features. No iPSC-derived neuron or organoid model of NEDASMB has been reported.
- **Applications:** Existing models are well suited to dissecting UBE3C's role in proteasome processivity and stress-responsive proteostasis, providing a mechanistic bridge to the human disorder.

Mechanistic Model / Interpretation

NEDASMB is best understood as a **"UBE3-family" neurodevelopmental proteostasis disorder**, sitting alongside its paralogs:

Gene	Disorder	Inheritance	Shared features
UBE3A	Angelman syndrome	Maternal deletion/ LoF (imprinted)	Severe ID, absent speech, movement disorder, seizures, behavioral phenotype
UBE3B	Kaufman oculocerebrofacial syndrome	Autosomal recessive	ID, developmental abnormalities
UBE3C	**NEDASMB (this disorder)**	**Autosomal recessive (biallelic LoF)**	**Severe DD/ID, absent speech, movement disorder, seizures, behavioral abnormalities**

The unifying biology is that **HECT-type UBE3 E3 ligases are required for cortical circuit assembly and higher cognition** [PMID: 33182779]. UBE3C's specific niche is the **proteasome**: it is a processivity-enhancing ligase [PMID: 28396413] that builds K48/K29 chains [PMID: 25752577] and, under stress, regulates proteasome capacity by ubiquitinating Rpn13 [PMID: 24811749]. Biallelic loss therefore degrades neuronal protein-quality-control capacity precisely when the developing brain is most dependent on it, yielding a static but severe encephalopathy. The paralog-directed discovery strategy — screening Angelman-like patients for *UBE3C* because it is *UBE3A*'s paralog — is itself strong circumstantial evidence for shared pathobiology and "explains some of the missing heritability" in Angelman-negative patients [PMID: 36401616].

Evidence Base

PMID	Title (abbrev.)	Role in this report
PMID: 36401616	*Biallelic variants in HECT E3 paralogs HECTD4 and UBE3C cause NDDs overlapping Angelman syndrome*	**Primary disease-defining paper.** Establishes gene, inheritance, LoF mechanism, phenotype, discovery strategy
PMID: 33182779	*Molecular evolution, neurodevelopmental roles and clinical significance of HECT-type UBE3 ligases*	Frames UBE3-family requirement for cortical development/cognition
PMID: 32039437	*Crystal structure of HECT domain of UBE3C E3 ligase*	Enzyme class, catalytic domain, autoubiquitination, cellular roles
PMID: 25752577	*Assembly and recognition of K29-/K33-linked polyubiquitin*	UBE3C chain-linkage specificity (K48/K29)
PMID: 28396413	*Ubiquitinated proteins promote proteasome association of Usp14 and Ube3c*	UBE3C/Hul5 as proteasome processivity factor
PMID: 24811749	*Autoubiquitination of 26S proteasome on Rpn13*	Stress-responsive substrate (Rpn13) and proteostasis regulation
PMID: 17848870	*Angelman syndrome revisited*	Defines the Angelman phenotype the disorder overlaps
PMID: 25752573	*K29-selective ubiquitin binding domain*	Confirms UBE3C generates K29 chains
PMID: 28425671	*Activity-based probes for HECT E3 ligases*	Confirms endogenous UBE3C catalytic activity
PMID: 21167755	*HECT E3 ligase RAUL regulates type-I IFN*	Context for HECT-ligase immune roles (not disease-specific)
PMID: 27255693	*MEF2C haploinsufficiency syndrome*	Differential diagnosis
PMID: 39275948	*ACTL6B-related brain disorders*	Differential diagnosis (severe DD/ID, absent speech)
PMID: 36293418	*CTNNB1 syndrome*	Differential diagnosis

PMID	Title (abbrev.)	Role in this report
PMID: 37927187	*KMT5B-related NDD*	Differential diagnosis
PMID: 38058756	*SYT1-related disorder*	Differential diagnosis (synaptopathy)
PMID: 31400086	*Mosaic Angelman syndrome*	Phenotypic spectrum context
PMID: 29710313	*Gaze/AAC in IDD*	Communication-support relevance

Limitations and Knowledge Gaps

1. **Tiny evidence base.** The *UBE3C*-specific phenotype rests on **three patients from two families** in a single 2023 report [PMID: 36401616]. All prevalence, penetrance, expressivity, sex ratio, and natural-history parameters are therefore **undefined**.
2. **No disease-specific omics.** There are no NEDASMB transcriptomic, proteomic, metabolomic, lipidomic, single-cell, or spatial datasets; the only patient functional data are RNA studies confirming the LoF effect.
3. **Variant classification lag.** ClinVar contains mostly VUS for isolated *UBE3C* SNVs; the disease alleles are private and not broadly deposited, complicating clinical interpretation.
4. **Mechanism is inferential.** The proteostasis → cortical-development causal chain is well supported for the UBE3 family and UBE3C biochemistry, but a **direct demonstration in patient-relevant neurons** (e.g., iPSC-derived) is lacking.
5. **No dedicated Orphanet/ICD code**, no formal diagnostic criteria, no QoL data, and no prognostic or treatment data specific to the disorder.
6. **Model gap.** No mouse model carries a patient-specific biallelic LoF allele evaluated against the full human phenotype; no organoid/iPSC model exists.

Proposed Follow-up Experiments / Actions

1. **Cohort expansion via GeneMatcher / Matchmaker Exchange** to identify additional biallelic *UBE3C* families, enabling estimation of penetrance, expressivity, natural history, and genotype–phenotype correlation.
2. **Patient iPSC-derived cortical neurons / cerebral organoids** to test the proteostasis hypothesis directly: measure proteasome processivity, K48/K29 ubiquitin-conjugate accumulation, and neuronal maturation/synaptic phenotypes in UBE3C-null vs. isogenic control.
3. **Deep-phenotype mouse *Ube3c*-null lines** (beyond IMPC screen) — cognition, seizure susceptibility (EEG), motor/cerebellar function, corpus-callosum morphometry — to establish a validated model.
4. **Functional variant classification pipeline** (e.g., ubiquitination assays with activity-based probes [PMID: 28425671]) to reclassify *UBE3C* VUS and support ClinVar submissions.
5. **Standardize the disorder in ontologies:** propose an Orphanet entry and ICD-11 mapping; enrich HPO annotations with frequency data as the cohort grows.
6. **Proteomic/ubiquitinome profiling** of patient-derived cells to define the UBE3C-dependent substrate landscape in neurons and nominate candidate biomarkers.
7. **Genetic-counseling resources** for affected families, including carrier-screening and reproductive-option (PGT-M/prenatal) guidance.

Report compiled from 7 confirmed findings, 17 reviewed papers, and ontology/database cross-mapping (OMIM, MONDO, MedGen, HGNC, UniProt, Reactome, KEGG, ClinVar, gnomAD, IMPC) across 5 investigation iterations.
